# EEE104 – Digital Electronics (I)

# Lecture 7

Dr. Yuxiao Fang

Emerging Electronics and Energy Technologies Research Centre (EEET)

Office: SC242

Office hour: 1:00-5:00 pm, Wednesday

# In This Session

- Boolean Algebra and Logic Simplification
  - Boolean Operations
  - Laws and Rules
  - DeMorgan's Theorems
  - Logic Simplification

# Boolean Operations

**Concepts**

- **Boolean algebra** is the mathematics of logic functions.
- A **variable** is a symbol used to represent a logical quantity. It can have a 1 or 0 value.
- The **complement** is the inverse of a variable and is indicated by an overbar, e.g. $\overline{A}$
- A **literal** is a variable or its complement.

**A variable and its complement are two literals of the same variable.**

# Boolean Operations

## Boolean Addition

Equivalent to the OR operation, with the basic rules:

| Inputs | OR result |
|---|---|
| **0 + 0** | **0** |
| 0 + 1 | 1 |
| 1 + 0 | 1 |
| 1 + 1 | 1 |

A sum term is a sum of literals, for example:

$$A + B + C + D$$

**A sum term is 0 only when every literal is 0.**

# Boolean Operations

## Boolean Multiplication

Equivalent to the AND operation, with the basic rules:

| Inputs | AND result |
|---|---|
| 0 · 0 | 0 |
| 0 · 1 | 0 |
| 1 · 0 | 0 |
| **1 · 1** | **1** |

A product term is a product of literals, for example:

$ABCD$

**A product term is 1 only when every literal is 1.**

# Laws of Boolean Algebra

**Commutative Laws**

$$A + B = B + A \qquad AB = BA$$

**Associative Laws**

$$A + (B + C) = (A + B) + C$$

$$A(BC) = (AB)C$$

**Distributive Law (or Factoring)**

$$A(B + C) = AB + AC$$

# Rules of Boolean Algebra

1. $A + 0 = A$
2. $A + 1 = 1$
3. $A \cdot 0 = 0$
4. $A \cdot 1 = A$
5. $A + A = A$
6. $A + \overline{A} = 1$
7. $A \cdot A = A$
8. $A \cdot \overline{A} = 0$
9. $\overline{\overline{A}} = A$
10. $\mathbf{A + AB = A}$
11. $\mathbf{A + \overline{A}B = A + B}$
12. $\mathbf{(A + B)(A + C) = A + BC}$

A, B and C may represent variables or whole expressions.

Rules 10-12 follow from simpler rules and the laws.

# Rules of Boolean Algebra

**Rule 10: Absorption**

$$\mathbf{A + AB = A}$$

$$A + AB = A(1 + B) = A$$

If A = 1, the OR output is already 1. If A = 0, AB = 0.

**Rule 11: Complemented absorption**

$$\mathbf{A + \overline{A}B = A + B}$$

$$A + \overline{A}B = A + AB + \overline{A}B$$

$$= A + (A + \overline{A})B = A + B$$

**Rule 11: when A = 0, the remaining output is B.**

# Rules of Boolean Algebra

**Rule 12: Product of two sums**

$$\mathbf{(A + B)(A + C) = A + BC}$$

$$(A + B)(A + C) = AA + AC + AB + BC$$

$$= A + AC + AB + BC$$

$$= A + AB + BC = A + BC$$

The shared term must be the same in both brackets.

**Practice**

1. $R + R(S + T)$

2. $(P + Q)(P + R) + P$

# Simplification Using Boolean Algebra

Simplify an expression to reduce the gates needed for implementation.

$$AB + A(B + C) + B(B + C)$$

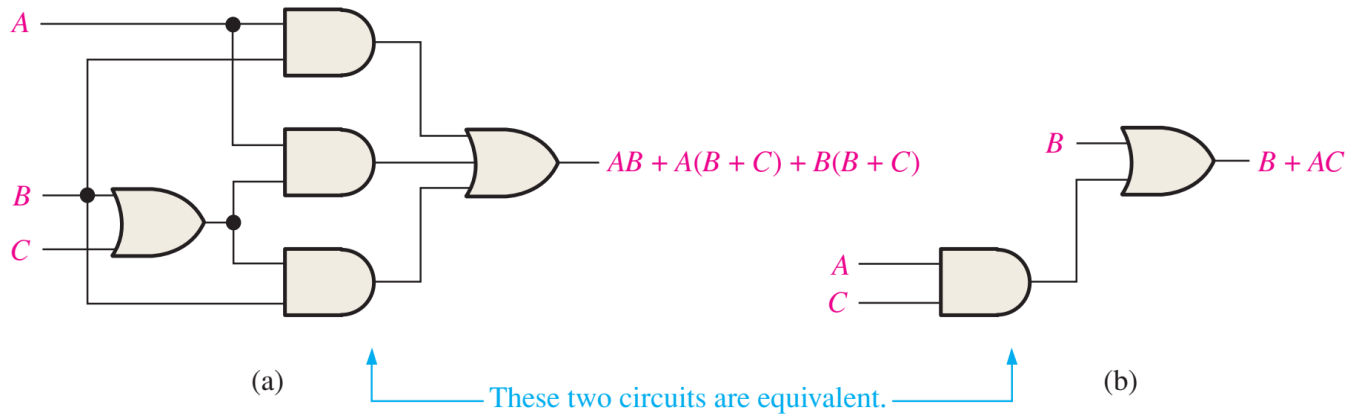
$$= AB + AB + AC + BB + BC$$

$$= AB + AC + B + BC$$

$$= \mathbf{B + AC}$$

**Rule 10: Absorption**

$$\mathbf{A + AB = A}$$

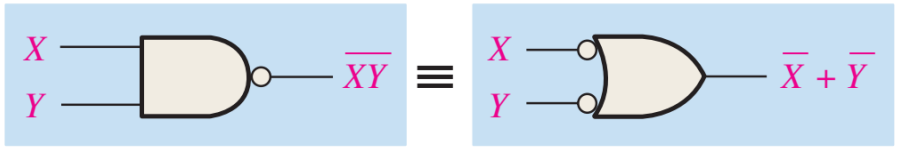


# DeMorgan's Theorems (1st)

The complement of a product equals the sum of the complements.

$$\overline{XY} = \overline{X} + \overline{Y}$$

| X | Y | Left side | Right side |
|---|---|---|---|
| 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |



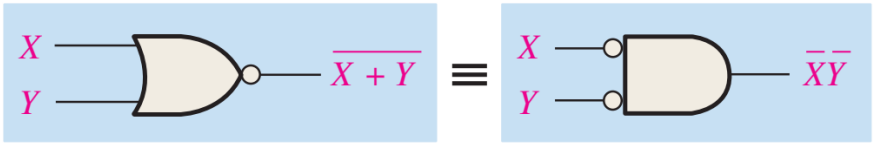

NAND Negative-OR

# DeMorgan's Theorems (2nd)

The complement of a sum equals the product of the complements.

$$\overline{X + Y} = \overline{X}\,\overline{Y}$$

| X | Y | Left side | Right side |
|---|---|---|---|
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 |



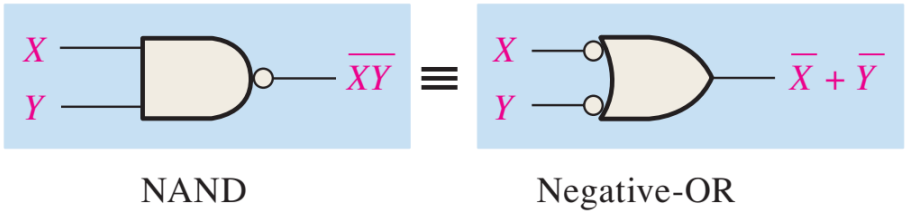

NOR Negative-AND

# DeMorgan's Theorems

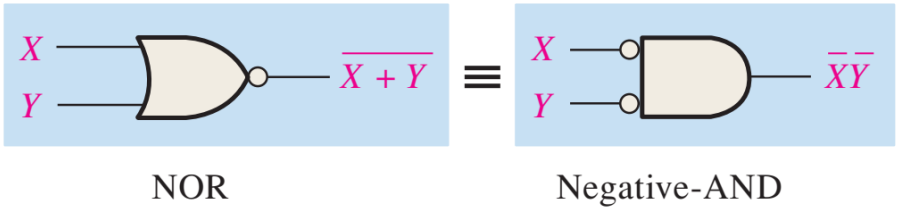
**Break the bar, change the sign.**

Work on the whole expression under the bar.

$$\overline{XY} = \overline{X} + \overline{Y}$$




$$\overline{X + Y} = \overline{X}\,\overline{Y}$$




**Practice**

$$\overline{(P + Q)R}$$

# DeMorgan's Theorems

## Application Examples

Each variable may represent a combination of other variables.

$$\overline{(AB + C)(A + BC)} = \overline{AB + C} + \overline{A + BC}$$

$$= \overline{AB}\,\overline{C} + \overline{A}\,\overline{BC}$$

$$= (\overline{A} + \overline{B})\overline{C} + \overline{A}(\overline{B} + \overline{C})$$

$$\overline{\overline{A + B\overline{C}} + \overline{DE + \overline{F}}} = \overline{\overline{A + B\overline{C}}}\;\overline{\overline{DE + \overline{F}}}$$

$$= (A + B\overline{C})(\overline{D} + \overline{\overline{E + \overline{F}}})$$

$$= (A + B\overline{C})(\overline{D} + E + \overline{F})$$